

Author: Jacky Leung

This is a concise, cheat-sheet-style set of notes for ECON 3143: Macroeconomic Theory II at HKUST. The primary focus is on the mathematical derivation of models, though key economic intuitions are also highlighted. Graphical examples, such as the movements of curves, are largely omitted for brevity. Content is based on Professor Jenny Xu's lecture notes and course materials, and the textbook *Intermediate Macroeconomics* by Garin, Lester, and Sims.

1 Introduction

- Macroeconomics concerns **growth** (long-run) and **cycles** (short-run).
- How can we isolate these two from raw data?
 - Decomposition: growth component $y_t^{\text{trend}} = \ln Y_t^{\text{trend}}$ and cyclical component $y_t = \ln Y_t^{\text{cycle}}$

$$\begin{aligned}\ln Y_t &= \ln Y_t^{\text{trend}} + \ln Y_t^{\text{cycle}} \\ y_t &= \ln Y_t - y_t^{\text{trend}}\end{aligned}$$

A mathematical property:

$$\begin{aligned}y_t &= \ln \left(\frac{Y_t}{Y_t^{\text{trend}}} \right) \\ &= \ln \left(\frac{Y_t - Y_t^{\text{trend}}}{Y_t^{\text{trend}}} + 1 \right) \\ &\approx \frac{Y_t - Y_t^{\text{trend}}}{Y_t^{\text{trend}}}\end{aligned}$$

1.1 Methods for decomposition

1. Linear Detrending:

Assume a linear growth in the long run:

$$\begin{aligned}Y_t^{\text{trend}} &= Y_0^{\text{trend}}(1 + g)^t \\ \ln Y_t^{\text{trend}} &= \ln Y_0^{\text{trend}} + t \ln(1 + g)\end{aligned}$$

Some problems under linear detrending:

- (1) Business cycles are persistent,
- (2) Cannot capture structural breaks (non-linear patterns).

2. Hodrick-Prescott (HP) Filter:

Instead of assuming y_t^{trend} to be linear, we define a regularized minimization problem:

$$\begin{aligned}\min_{y_t, y_t^{\text{trend}}} \quad & \sum_{t=1}^T (\ln(Y_t) - y_t^{\text{trend}})^2 + \lambda \sum_{t=1}^T [(y_{t+1}^{\text{trend}} - y_t^{\text{trend}}) - (y_t^{\text{trend}} - y_{t-1}^{\text{trend}})]^2 \\ \text{s.t.} \quad & y_t + y_t^{\text{trend}} = \ln Y_t\end{aligned}$$

- If λ is too small, the smoothing factor is insignificant $\implies$ **Overfitting**.
- If λ is too big, the smoothing effect is too large $\implies$ **Underfitting**.
 - If $\lambda \rightarrow \infty$, we have $y_{t+1}^{\text{trend}} - y_t^{\text{trend}} = y_t^{\text{trend}} - y_{t-1}^{\text{trend}} \implies y_t^{\text{trend}} = \frac{y_{t-1}^{\text{trend}} + y_{t+1}^{\text{trend}}}{2}$, which is just the linear trend.
- For high frequency data, the trend is less likely to change across periods $\implies$ higher λ for regularization.
- Rule of thumb: $\lambda = 1600$ for quarterly data, $\lambda = 100$ for annual data.

2 Economic Growth and the Solow Model

2.1 Facts about economic growth

- The **Kaldor's facts** about domestic growth:
 1. Output per worker grows over time, and it does not tend to diminish: $\frac{Y_t}{N_t} \uparrow$
 2. Capital per worker grows over time: $\frac{K_t}{N_t} \uparrow$
 3. Capital-output ratio is approximately constant: $\frac{K_t}{Y_t} \sim$
 4. The share of labor in national income is nearly constant: $\frac{w_t N_t}{Y_t} \sim$
 5. Rate of return to capital is nearly constant: $\text{MPK} = \frac{\partial K_t}{\partial Y_t} \sim$
 6. Real wage grows at a sustained, nearly constant rate: $w_t \uparrow$
- More facts on international level:
 7. There are enormous variations in income across countries.
 8. There are growth miracles and growth disasters.
 9. Empirical evidence on convergence:
 - **Absolute convergence:** Poor countries should grow faster (per capita) than rich countries without conditioning on any other factors.
 - * Growth rate of real GDP per capita should be negatively related to the starting level:

$$\frac{Y_{it} - Y_{i0}}{Y_{i0}} \approx \ln Y_{it} - \ln Y_{i0} = \beta_0 + \beta_1 \ln Y_{i0} + \epsilon_i, \beta_1 < 0 \text{ if absolute convergence holds}$$

- * Empirically holds in homogeneous countries (e.g. OECD), but not in broader samples.
- **Conditional convergence:** Poor countries should grow faster (per capita) than rich countries, conditional on country-specific factors.

$$\ln Y_{it} - \ln Y_{i0} = \beta_0 + \beta_1 \ln Y_{i0} + \beta^T \mathbf{x}_{i0} + \epsilon_i, \beta_1 < 0 \text{ if absolute convergence holds}$$

- * Empirically holds, and the rate of convergence is about 2% per year.
- * We can numerically solve for:

$$\frac{y^* - y_t}{y^* - y_0} = e^{-\lambda t} = 0.5$$

If $\lambda = 0.02$, we have $t \approx 35$. That is, it takes about 35 years to close half of the gap between the initial level and the target level of real GDP per capita.

- We want to develop a model of growth that is consistent with these facts.

2.2 The Solow Model

- Assumptions:
 - Given technology Z_t , the representative firm chooses capital K_t and labor N_t to produce output Y_t to maximize profit π_t .
 - The representative household owns capital and labor. Furthermore, household has an exogenous constant saving rate s . Household does not optimize.
 - Closed economy. Output = income. Investment = saving.
 - Exogenous growth in productivity and population:

$$N_{t+1} = (1+n)N_t, Z_{t+1} = (1+z)Z_t$$

- Production function:

$$Y_t = F(K_t, Z_t N_t)$$

- Constant return to scale (CRTS): $F(\lambda K_t, \lambda Z_t N_t) = \lambda F(K_t, Z_t N_t)$.
- Diminishing marginal product: $\frac{\partial Y_t}{\partial K_t} > 0, \frac{\partial Y_t}{\partial Z_t N_t} > 0, \frac{\partial^2 Y_t}{\partial K_t^2} < 0, \frac{\partial^2 Y_t}{\partial (Z_t N_t)^2} < 0$.
- Inada conditions: $\lim_{k \rightarrow 0} \frac{\partial Y_t}{\partial K_t} = \infty, \lim_{k \rightarrow \infty} \frac{\partial Y_t}{\partial K_t} = 0$.
- Example - Cobb-Douglas: $Y_t = K^\alpha (Z_t N_t)^{1-\alpha}$.

- Intensive form of production function:

- Since the production function is CRTS, dividing both sides by $Z_t N_t$ yields:

$$\frac{Y_t}{Z_t N_t} = F\left(\frac{K_t}{Z_t N_t}, 1\right)$$

$$\hat{y}_t = f(\hat{k}_t)$$

- Example - Cobb-Douglas: $\hat{y}_t = \hat{k}_t^\alpha$.

- Firm's decision:

- Given wage rate w_t and rent R_t , the firm chooses K_t and N_t to maximize π_t :

$$\max_{K_t, N_t} \pi_t = Y_t - R_t K_t - w_t N_t$$

- Taking FOCs:

$$\frac{\partial \pi_t}{\partial K_t} = \frac{\partial Y_t}{\partial K_t} - R_t = 0 \implies \frac{\partial Y_t}{\partial K_t} = \text{MPK} = R_t$$

$$\frac{\partial \pi_t}{\partial N_t} = \frac{\partial Y_t}{\partial N_t} - w_t = 0 \implies \frac{\partial Y_t}{\partial N_t} = \text{MPL} = w_t$$

- For a CRTS production function, we can actually get $\pi_t^* = 0$.
- Example - Cobb-Douglas (derivation is omitted):

$$R_t = \alpha \hat{k}_t^{\alpha-1}$$

$$w_t = Z_t (1 - \alpha) \hat{k}_t^\alpha$$

$$\frac{w_t N_t}{Y_t} = 1 - \alpha$$

- We will show later that $\hat{k}_t$ is constant over time, so these three equations are consistent with Facts (5), (6), and (4) respectively.

- Household's decision:

- Budget constraint:

$$C_t + I_t = w_t N_t + R_t K_t + \pi_t$$

Since $\pi_t^* = 0$,

$$C_t + I_t = w_t N_t + R_t K_t = Y_t$$

- Savings:

$$I_t = s Y_t, \quad C_t = (1 - s) Y_t$$

- Capital accumulation equation given depreciation rate δ :

$$K_{t+1} = I_t + (1 - \delta) K_t$$

- We want to derive the intensive form of the capital accumulation equation to analyze the dynamics of capitals.

$$K_{t+1} = sF(K_t, Z_t N_t) + (1 - \delta)K_t$$

Dividing both sides by $Z_t N_t$ and rearranging some terms:

$$\frac{K_{t+1}}{Z_{t+1} N_{t+1}} \cdot \frac{Z_{t+1}}{Z_t} \cdot \frac{N_{t+1}}{N_t} = sf(\hat{k}_t) + (1 - \delta)\hat{k}_t$$

Which gives us:

$$(1 + n)(1 + z)\hat{k}_{t+1} = sf(\hat{k}_t) + (1 - \delta)\hat{k}_t$$

Finally:

$$\hat{k}_{t+1} = \left(\frac{1}{(1 + n)(1 + z)} \right) (sf(\hat{k}_t) + (1 - \delta)\hat{k}_t)$$

- At the **steady state**, $\hat{k}_{t+1} = \hat{k}_t = \hat{k}^*$.

$$\begin{aligned} \frac{K_{t+1}}{Z_{t+1} L_{t+1}} &= \frac{K_t}{Z_t L_t} \\ \frac{K_{t+1}}{K_t} &= \frac{Z_{t+1} L_{t+1}}{Z_t L_t} = (1 + n)(1 + z) \end{aligned}$$

- The production function is CRTS; if both K and ZN grow at the same rate $(1 + n)(1 + z)$, Y and C must grow at this rate as well.
- The growth of per-capita variables:

$$\frac{\frac{K_{t+1}}{N_{t+1}}}{\frac{K_t}{N_t}} = \frac{\frac{K_{t+1}}{K_t}}{\frac{N_{t+1}}{N_t}} = 1 + z$$

- Similarly, $\frac{K}{N}$, $\frac{Y}{N}$, $\frac{C}{N}$ all grow at the same rate $1 + z$.
- Consistent with Facts (1), (2) and (3).
- Solving for $\hat{k}^*$:

$$\begin{aligned} \hat{k}^* &= \left(\frac{1}{(1 + n)(1 + z)} \right) (sf(\hat{k}^*) + (1 - \delta)\hat{k}^*) \\ &= \left(\frac{1}{(1 + n)(1 + z) - (1 - \delta)} \right) (sf(\hat{k}^*)) \\ &\approx \left(\frac{1}{n + z + \delta} \right) (sf(\hat{k}^*)) \end{aligned}$$

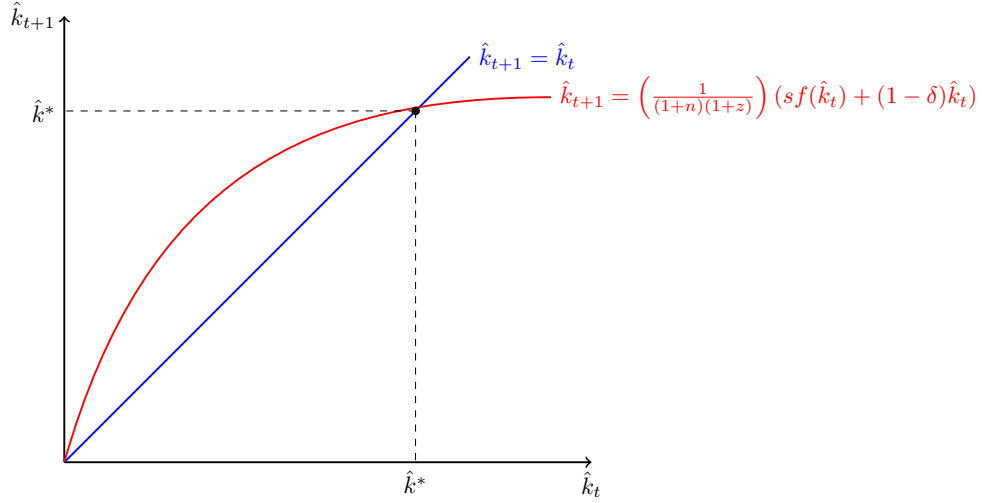
- Example - Cobb-Douglas: $\hat{k}^* = \left(\frac{s}{(1+n)(1+z)-(1-\delta)} \right)^{\frac{1}{1-\alpha}} \approx \left(\frac{s}{n+z+\delta} \right)^{\frac{1}{1-\alpha}}$.

- Graphical analysis (1): Plotting $\hat{k}_{t+1}$ against $\hat{k}_t$

- What is the shape of the curve?

$$\frac{d\hat{k}_{t+1}}{d\hat{k}_t} = \left(\frac{1}{(1 + n)(1 + z)} \right) (sf'(\hat{k}_t) + (1 - \delta))$$

- Since $f(\hat{k}_t)$ is concave, $\hat{k}_{t+1}(\hat{k}_t)$ is also concave.

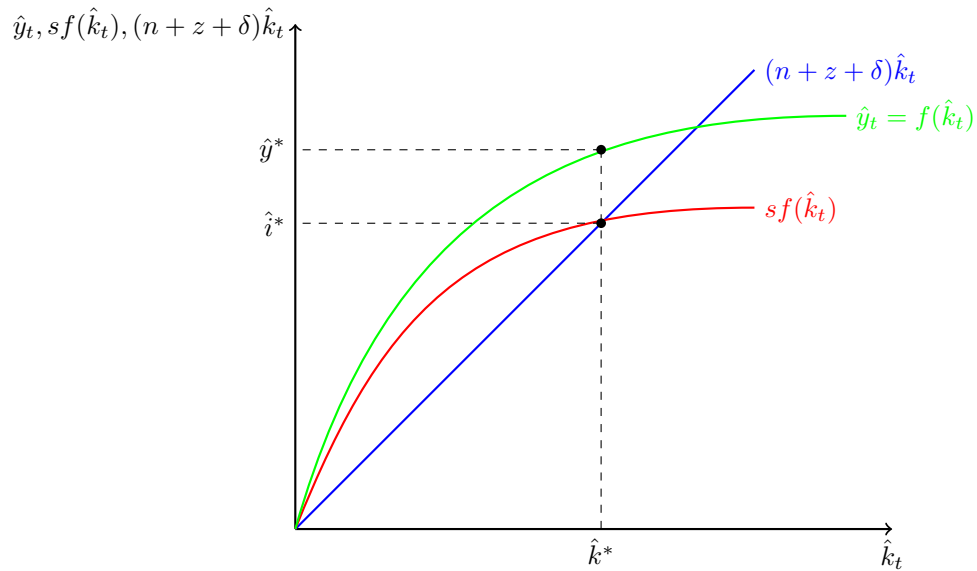


- If $\hat{k}_t > \hat{k}^*$, then $\hat{k}_{t+1} < \hat{k}_t$, so $\hat{k}$ decreases towards $\hat{k}^*$ over time, and vice versa.
- Mathematically:

$$\begin{aligned}
 g_{\hat{k}} &= \frac{\hat{k}_{t+1} - \hat{k}_t}{\hat{k}_t} \\
 &= \frac{s}{(1+n)(1+z)} \cdot \frac{f(\hat{k}_t)}{\hat{k}_t} + \frac{(1-\delta) - (1+n)(1+z)}{(1+n)(1+z)} \\
 &\approx \frac{s}{(1+n)(1+z)} \cdot \frac{f(\hat{k}_t)}{\hat{k}_t} - \frac{n+z+\delta}{(1+n)(1+z)}
 \end{aligned}$$

- Graphical analysis (2): Plotting $\hat{y}_t$ against $\hat{k}_t$

- We can make use of the equation $\hat{k}^* \approx \left(\frac{1}{n+z+\delta}\right) (sf(\hat{k}^*)) \implies sf(\hat{k}^*) \approx (n+z+\delta)\hat{k}^*$ derived previously to construct the graph:



- Using these graphs, we can easily analyze the dynamics when any of the exogenous variable (s, n, z, δ) changes. (Exercise: what if s increases?)
- Golden rule for saving rate:
 - What is the optimal s that maximizes consumption?
 - Again, we make use of the equation $sf(\hat{k}^*) \approx (n + z + \delta)\hat{k}^*$.

$$\begin{aligned}\hat{c}^* &= (1 - s)f(\hat{k}^*) \\ &= f(\hat{k}^*) - (n + z + \delta)\hat{k}^*\end{aligned}$$

Taking FOC:

$$\frac{\partial \hat{c}^*}{\partial s} = (f'(\hat{k}^*) - (n + z + \delta)) \frac{\partial \hat{k}^*}{\partial s} = 0$$

Since $\frac{\partial \hat{k}^*}{\partial s} > 0$ always, the term $f'(\hat{k}^*) - (n + z + \delta)$ must equal 0.

$$f'(\hat{k}^*) = n + z + \delta$$

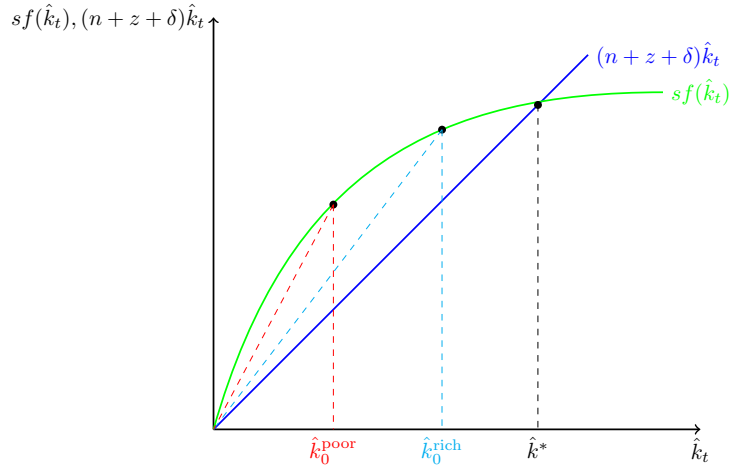
Example - Cobb-Douglas:

$$\begin{aligned}\alpha(\hat{k}^*)^{\alpha-1} &= n + z + \delta \\ \alpha \left(\left(\frac{s}{n + z + \delta} \right)^{\frac{1}{1-\alpha}} \right)^{\alpha-1} &= n + z + \delta \\ s^{\text{gr}} &= \alpha\end{aligned}$$

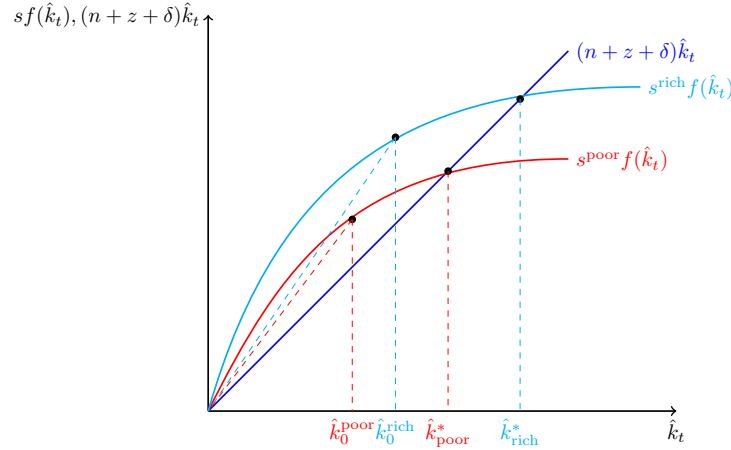
- Problem: Because we assume household do not optimize, the model allows $s \neq s^{\text{gr}} \implies$ dynamic inefficiency.
- Does the model explain cross-country convergence?
 - Recall that:

$$g_{\hat{k}} \approx \frac{s}{(1+n)(1+z)} \cdot \frac{f(\hat{k}_t)}{\hat{k}_t} - \frac{n+z+\delta}{(1+n)(1+z)}$$

If two countries have the same s, n, z, δ but $\hat{k}_t^{\text{poor}} < \hat{k}_t^{\text{rich}}$, then $g_{\hat{k}}^{\text{poor}} > g_{\hat{k}}^{\text{rich}}$, and furthermore they have the same $\hat{k}^*$.



- The slope of the hypotenuse $= \frac{sf(\hat{k}_t)}{\hat{k}_t}$ tells us about $g_{\hat{k}}$, and clearly the poor economy has a steeper slope and thus a higher growth rate.
- This aligns with conditional convergence.
- What if the two countries have different s ?



- In this case, the rich country has higher growth rate.
- Rejects absolute convergence.
- Although the model explains the facts about convergence qualitatively, empirical data shows that the predicted convergence rate is too fast.
- Does the model explain cross-country different in output per capita?
 - Difference in $\frac{K}{N}$?
 - * Consider $\tilde{y}_1 = \frac{Y_1}{N_1} = \left(\frac{K_1}{N_1}\right)^\alpha Z^{1-\alpha}$ and $\tilde{y}_2 = \frac{Y_2}{N_2} = \left(\frac{K_2}{N_2}\right)^\alpha Z^{1-\alpha}$.
 - * If $\frac{\tilde{y}_1}{\tilde{y}_2} = X$, then their $\frac{K}{N}$ must differ by $X^{\frac{1}{\alpha}}$. If $X = 10$ and $\alpha = \frac{1}{3}$, the difference in $\frac{K}{N}$ must be 1000 times.
 - * Not plausible.
 - Difference in s ?
 - * Textbook exercise.
 - Difference in Z ?
 - How to measure technology / productivity exactly?

$$Y_t = A_t K_t^\alpha (Z_t N_t)^{1-\alpha}$$

$$\ln Y_t = \ln A_t + \alpha \ln K_t + (1-\alpha) \ln Z_t + (1-\alpha) \ln N_t$$

Define total factor productivity (TFP) as the proportion of income growth that cannot be attributed to K_t or N_t .

$$\ln \text{TFP}_t = \ln A_t + (1-\alpha) \ln Z_t = \ln Y_t - \alpha \ln K_t - (1-\alpha) \ln N_t$$

- Young (1992): Growth in HK can be explained by growth in technology, but not in Singapore.
- Still, correlation between GDP and TDP is high in broader sample.
- Solow model predicts sustained growth must be driven by productivity Z_t , but does not explain what productivity is.
- Some factors affecting productivity: Knowledge and education, Climate, Geography, Institution, Finance, Degree of openness
- Policy implication: Focus on policies that promote productivity.

3 Consumption

- Previously we assumed consumer do not choose savings to optimize their consumption.
- We can use a simple two-period model to analyze consumption behavior.
- Assumptions:
 - Two periods: t and $t + 1$.
 - Exchange economy. No money, no production, only endowment.
 - A representative consumer.

- Budget constraints:

- Period t :

$$C_t + S_t = Y_t$$

- Period $t + 1$:

$$C_{t+1} + S_{t+1} = Y_{t+1} + (1 + r)S_t$$

Household would not allow $S_{t+1} > 0$ as there is no next period to consume. Lenders would not lend for $S_{t+1} < 0$ as the household would not repay. Thus the terminal condition is $S_{t+1} = 0$.

$$C_{t+1} = Y_{t+1} + (1 + r_t)S_t$$

- Combining the two equations by letting S_t equal each other, we have the intertemporal budget constraint:

$$C_t + \frac{C_{t+1}}{1 + r_t} = Y_t + \frac{Y_{t+1}}{1 + r_t}$$

- PDV of consumption = PDV of income.

- Utility:

- $u'(C) > 0, u''(C) < 0$.
- Total utility: $U(C_t, C_{t+1}) = u(C_t) + \beta u(C_{t+1})$, where $0 < \beta < 1$.

- The maximization problem:

$$\max_{C_t, C_{t+1}} U(C_t, C_{t+1}) = u(C_t) + \beta u(C_{t+1})$$

$$\text{s.t. } C_t + \frac{C_{t+1}}{1 + r_t} = Y_t + \frac{Y_{t+1}}{1 + r_t}$$

Setting up the Lagrangian:

$$\mathcal{L}(C_t, C_{t+1}, \lambda) = u(C_t) + \beta u(C_{t+1}) + \lambda(Y_t + \frac{Y_{t+1}}{1 + r_t} - C_t - \frac{C_{t+1}}{1 + r_t})$$

Taking FOCs:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial C_t} = u'(C_t) - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial C_{t+1}} = \beta u'(C_{t+1}) - \frac{\lambda}{1 + r_t} = 0 \end{cases}$$

Setting λ equal each other, we can get the Euler equation:

$$u'(C_t) = \beta(1 + r_t)u'(C_{t+1})$$

The intuition: MB = MC of present consumption.

- Substituting the budget constraint $C_{t+1} = (1 + r_t)Y_t + Y_{t+1} - (1 + r_t)C_t$ into the Euler equation, we can find the consumption function:

$$C_t = C^d(Y_t, Y_{t+1}, r_t)$$

- Example - log utility: $u(C) = \ln C$.

- The Euler equation:

$$\frac{1}{C_t} = \beta(1 + r_t) \frac{1}{C_{t+1}} \implies C_{t+1} = \beta(1 + r_t)C_t$$

Hence,

$$\begin{aligned} \beta(1 + r_t)C_t &= (1 + r_t)Y_t + Y_{t+1} - (1 + r_t)C_t \\ C_t &= \left(\frac{1}{1 + \beta} \right) \left(Y_t + \frac{Y_{t+1}}{1 + r_t} \right) \end{aligned}$$

- The indifference curve:

- Recall that $U = u(C_t) + \beta u(C_{t+1})$.
- To find the slope of the indifference curve, a shortcut is using the implicit function theorem:

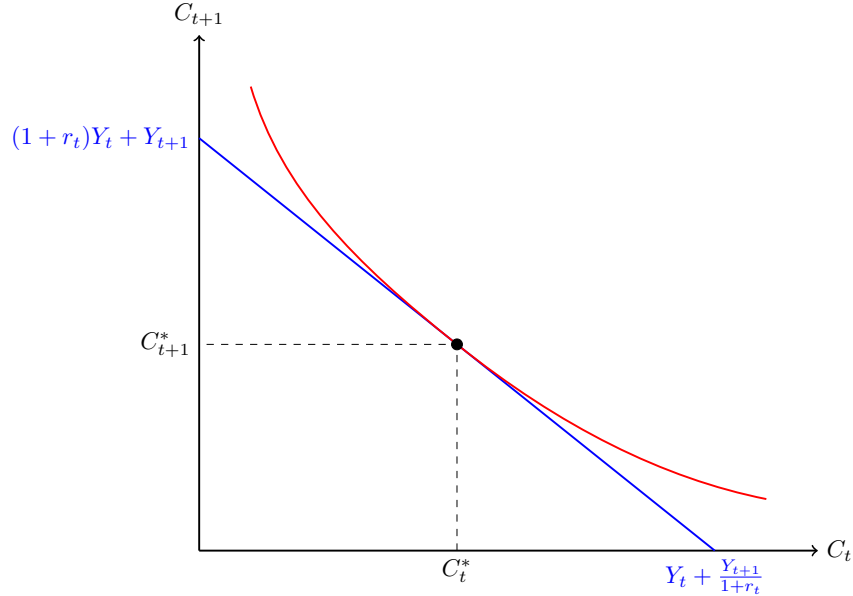
$$\frac{dC_{t+1}}{dC_t} = - \frac{\frac{\partial U}{\partial C_t}}{\frac{\partial U}{\partial C_{t+1}}} = - \frac{u'(C_t)}{\beta u'(C_{t+1})}$$

At the optimum consumption point, by the Euler equation, $u'(C_t) = \beta(1 + r_t)u'(C_{t+1})$. Thus, at the point, the slope to the indifference curve is:

$$\frac{dC_{t+1}}{dC_t} = -(1 + r_t)$$

Which we can see that the budget line exactly tangents to the point:

$$C_{t+1} = (1 + r_t)Y_t + Y_{t+1} - (1 + r_t)C_t$$



- The graphical analysis of dynamics is omitted, but here we discuss the result verbally.

- Increase in Y_t or Y_{t+1} :
 - Given an **income effect**, since both C_t and C_{t+1} are normal goods, both C_t^* and C_{t+1}^* should increase.
 - **Consumption smoothing**: If the income in **either** period increases, the consumer will consume the increase across **both** periods.
 - $0 < \text{MPC} < 1$.
- Increase in r_t :
 - An important note: the consumption bundle (Y_t, Y_{t+1}) lies on the budget line.
 - Then, we can classify the consumer into two types.
 - Saver: $C_t^* < Y_t$
 - * Substitution effect: $C_t \downarrow, C_{t+1} \uparrow$
 - * Income effect: $C_t \uparrow, C_{t+1} \uparrow$
 - * Overall: $C_t?, C_{t+1} \uparrow$
 - Borrower: $C_t^* > Y_t$
 - * Substitution effect: $C_t \downarrow, C_{t+1} \uparrow$
 - * Income effect: $C_t \downarrow, C_{t+1} \downarrow$
 - * Overall: $C_t \downarrow, C_{t+1}?$
- Oftentimes we will assume substitution effect dominates $\implies \frac{\partial C_t}{\partial r_t} < 0$.
- Revisiting the log utility example, where $C_t = \left(\frac{1}{1+\beta}\right) \left(Y_t + \frac{Y_{t+1}}{1+r_t}\right)$.

$$\frac{\partial C_t}{\partial Y_t} = \frac{1}{1+\beta} < 1$$

$$\frac{\partial C_t}{\partial Y_{t+1}} = \frac{1}{(1+\beta)(1+r_t)} < 1$$

$$\frac{\partial C_t}{\partial r_t} = -\left(\frac{1}{1+\beta}\right) \left(\frac{Y_{t+1}}{(1+r_t)^2}\right) < 0$$

- Permanent income hypothesis: Consumption is a function of the present discounted value of the lifetime income.
 - Holds for the log utility case.

- When Y_t increases,

$$dC_t = \frac{\partial C^d(\cdot)}{\partial Y_t} dY_t + \frac{\partial C^d(\cdot)}{\partial Y_{t+1}} dY_{t+1} + \frac{\partial C^d(\cdot)}{\partial r_t} dr_t$$

- If the increase is a transitory income ($dY_t > 0, dY_{t+1} = 0$):

$$\frac{dC_t}{dY_t} = \frac{\partial C^d(\cdot)}{\partial Y_t}$$

- If the increase is a permanent income ($dY_t = dY_{t+1} > 0$):

$$\frac{dC_t}{dY_t} = \frac{\partial C^d(\cdot)}{\partial Y_t} + \frac{\partial C^d(\cdot)}{\partial Y_{t+1}}$$

- What if Y_{t+1} is uncertain?

- Let Y_{t+1} be a random variable.

$$Y_{t+1} = \begin{cases} Y_{t+1}^H = Y_t + \Delta & \text{with probability } p \\ Y_{t+1}^L = Y_t - \Delta & \text{with probability } 1 - p \end{cases}$$

- The expression for the intertemporal budget constraint remains the same, except that Y_{t+1} and C_{t+1} are now random variables.
- Recall the fact that $\mathbb{E}(u(X)) = \sum p_i u(x_i)$, and thus $\mathbb{E}(u'(X)) = \sum p_i u'(x_i)$.
- Solve for the maximization problem:

$$\max_{C_t, C_{t+1}} U(C_t, C_{t+1}) = u(C_t) + \beta \mathbb{E}(u(C_{t+1}))$$

$$\text{s.t. } C_t + \frac{C_{t+1}}{1 + r_t} = Y_t + \frac{Y_{t+1}}{1 + r_t}$$

Setting up the Lagrangian:

$$\mathcal{L}(C_t, C_{t+1}, \lambda) = u(C_t) + \beta \mathbb{E}(u(C_{t+1})) + \lambda(Y_t + \frac{Y_{t+1}}{1 + r_t} - C_t + \frac{C_{t+1}}{1 + r_t})$$

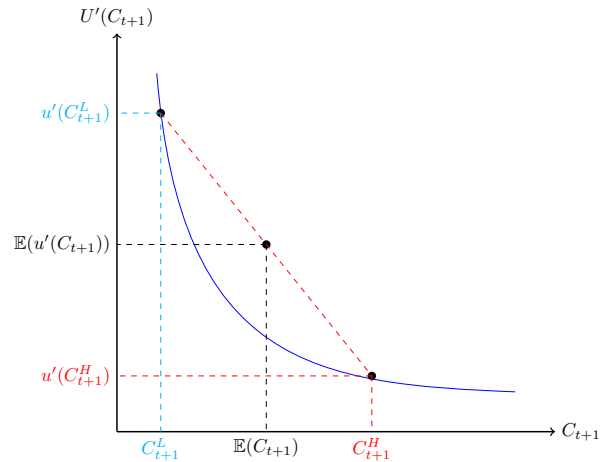
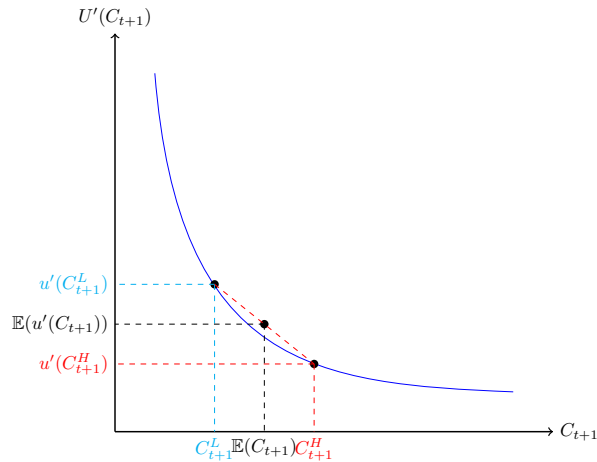
Taking FOCs:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial C_t} = u'(C_t) - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial C_{t+1}} = \beta \mathbb{E}(u'(C_{t+1})) - \frac{\lambda}{1 + r_t} = 0 \end{cases}$$

Setting λ equal each other, we can get the Euler equation:

$$u'(C_t) = \beta(1 + r_t) \mathbb{E}(u'(C_{t+1}))$$

- Since $\mathbb{E}(u'(C_{t+1})) \neq u'(\mathbb{E}(C_{t+1}))$ most of the time (Jensen's inequality), this is difficult to solve.
- For example, for log utility, $\mathbb{E}\left(\frac{1}{C_{t+1}}\right) \neq \frac{1}{\mathbb{E}(C_{t+1})}$.
- Special case: Quadratic utility.
- Will people save more if future income is more uncertain? (Precautionary saving)
- Yes, if $u'(C)$ is convex $\iff u'''(C) > 0$.



- The higher the variance, the higher $\mathbb{E}(u'(C_{t+1}))$. By the Euler equation, $u'(C_t)$ is higher, and because $u'(C_t)$ is strictly decreasing with C_t , C_t becomes lower and people save more instead.

- Intuition: MC of consumption $= u'(C_t)$ is high.
- For example, consumption was low after the Great Recession due to high future uncertainty.
- Mathematical property: mean-preserving spread ($u'(C_{t+1}^L)$ increases much faster than $u'(C_{t+1}^H)$ decreases).
- Therefore, consumers have two motives to save or borrow: consumption smoothing ($u''(C) < 0$) and risk aversion ($u'''(C) > 0$).
- Random walk hypothesis:
 - Without expectation, if $\beta = \frac{1}{1+r}$, then $u'(C_t) = u'(C_{t+1}) \implies C_t = C_{t+1}$.
 - With expectation, $u'(C_t) = \mathbb{E}(u'(C_{t+1})) \implies C_t = \mathbb{E}(C_{t+1})$ only if $u'''(C) = 0$.

4 General Equilibrium

- While we just assumed r_t to be exogenous, in an economy with many agents, r_t is actually determined endogenously by clearing the borrowing and lending market.
- First, we assume there are many households with the same preference. Each of the households, indexed by j , maximize their own utility by the standard formulation:

$$\begin{aligned} \max_{C_{j,t}, C_{j,t+1}} \quad & u(C_{j,t}) + \beta u(C_{j,t+1}) \\ \text{s.t.} \quad & C_{j,t} + \frac{C_{j,t+1}}{1+r_t} = Y_{j,t} + \frac{Y_{j,t+1}}{1+r_t} \end{aligned}$$

Then, we can obtain:

$$C_{j,t} = C^d(Y_{j,t}, Y_{j,t+1}, r_t)$$

- All markets clear, since every borrower must be matched with a reciprocal lender, the condition for aggregate savings S_t must be:

$$S_t = \sum S_{j,t} = 0$$

- Then,

$$\sum Y_{j,t} = \sum C_{j,t}$$

Which gives us a nice relationship between aggregate income Y_t and consumption C_t :

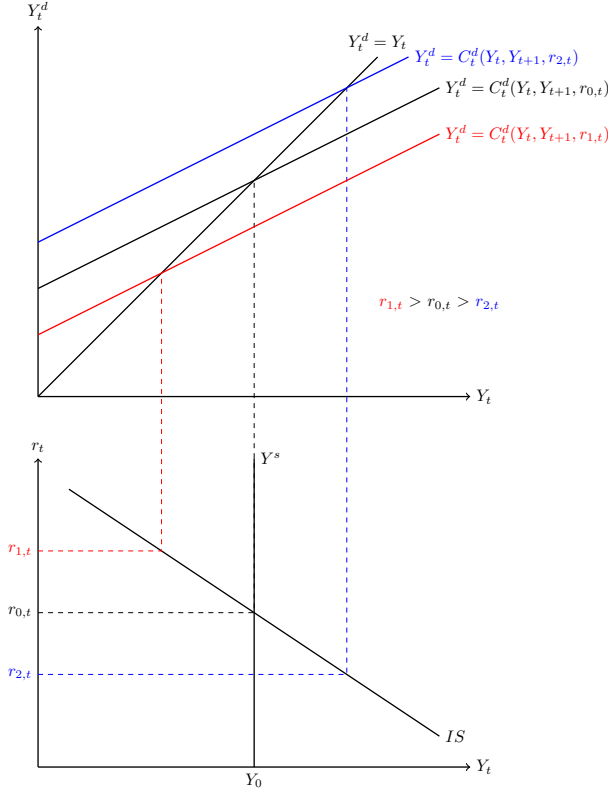
$$Y_t = C_t = C_t^d(Y_t, Y_{t+1}, r_t)$$

We can then use this to find the equilibrium r_t .

- Example - log utility:

$$\begin{aligned} Y_t &= \frac{1}{1+\beta} \left(Y_t + \frac{Y_{t+1}}{1+r_t} \right) \\ r_t &= \frac{1}{\beta} \cdot \frac{Y_{t+1}}{Y_t} \end{aligned}$$

- Note that we only assume everyone has the same preference. If households have different endowments, the conditions will still hold.
- This is the market demand, what about the supply?
- In an endowment economy, there is no production and Y_t is exogenously given. The supply is just a vertical line.
- Graphical representation:
 - Assume substitution effect dominates, so $C_t^d(Y_t, Y_{t+1}, r_t)$ is decreasing in r_t .
 - The slope $= \frac{\partial C_t^d}{\partial Y_t} = \text{MPC}$, $0 < \text{MPC} < 1$.



5 Production Economy

- Now, we can introduction firms to move from an endowment economy to a production economy.
- Firm:
 - We still use a CRTS and diminishing (e.g. Cobb-Douglas) production function, but not labor augmenting (i.e. no Z_t) for simplicity.

$$Y_t = A_t F(K_t, N_t)$$

- Capital accumulation equation:

$$K_{t+1} = I_t + (1 - \delta)K_t$$

In other words, if the firm simply chooses K_{t+1} , investment is implicitly defined by:

$$I_t = K_{t+1} - (1 - \delta)K_t$$

- The firm finances investment through borrowing with a premium f_t , so the cost of borrowing is:

$$r'_t = r_t + f_t$$

- Dividends:

- * Period t : The dividend is simply the profit: income minus labor cost.

$$D_t = Y_t - w_t N_t$$

- * Period $t + 1$: In addition to profits, since the firm ceases to exist, it liquidates the remaining capital. Also, it needs to repay the cost of debt.

$$D_{t+1} = Y_{t+1} + (1 - \delta)K_{t+1} - w_{t+1}N_{t+1} - (1 + r'_t)I_t$$

- The firm maximizes the firm value, which is the PDV of dividends, by choosing N_t and K_{t+1} . Note that all other variables, including K_t , are exogenously given. Substituting everything:

$$\begin{aligned}\max_{N_t, K_{t+1}} V_t &= D_t + \left(\frac{1}{1+r_t} \right) D_{t+1} \\ &= A_t F(K_t, N_t) - w_t N_t + \left(\frac{1}{1+r_t} \right) (A_{t+1} F(K_{t+1}, N_{t+1}) + (1-\delta)K_{t+1} \\ &\quad - w_{t+1} N_{t+1} - (1+r'_t)(K_{t+1} - (1-\delta)K_t))\end{aligned}$$

- Taking FOCs gives us the labor and investment demands respectively.
- Labor demand:

$$\begin{aligned}\frac{\partial V_t}{\partial N_t} &= A_t F_N(K_t, N_t) - w_t = 0 \\ w_t &= A_t F_N(K_t, N_t)\end{aligned}$$

- * The effects on N_t if each of the variables moves:
- * $w_t \uparrow \implies F_N(K_t, N_t) \uparrow \implies N_t \downarrow$
- * $A_t \uparrow \implies F_N(K_t, N_t) \downarrow \implies N_t \uparrow$
- * $K_t \uparrow, F_N(K_t, N_t) \sim \implies N_t \downarrow$
- * Therefore, the labor demand function (with symbols for relations) is:

$$N_t = N_t^d(w_t^-, A_t^+, K_t^-)$$

- Investment demand:

$$\begin{aligned}\frac{\partial V_t}{\partial K_{t+1}} &= \left(\frac{1}{1+r_t} \right) (A_{t+1} F_K(K_{t+1}, N_{t+1}) + (1-\delta) - (1+r'_t)) = 0 \\ 1+r'_t &= A_{t+1} F_K(K_{t+1}, N_{t+1}) + (1-\delta)\end{aligned}$$

- * Intuition: MB = MC of investment.
- * Further rearranging:

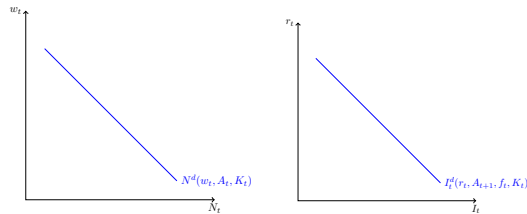
$$r_t + f_t + \delta = A_{t+1} F_K(K_{t+1}, N_{t+1})$$

- * Recall:

$$I_t = K_{t+1} - (1-\delta)K_t$$

- * The effects on I_t if each of the variables moves:
- * $r_t \uparrow \implies F_K(K_{t+1}, N_{t+1}) \uparrow \implies K_{t+1} \downarrow \implies I_t \downarrow$
- * $f_t \uparrow \implies F_K(K_{t+1}, N_{t+1}) \uparrow \implies K_{t+1} \downarrow \implies I_t \downarrow$
- * $A_{t+1} \uparrow \implies F_N(K_t, N_t) \downarrow \implies K_{t+1} \uparrow \implies I_t \uparrow$
- * $K_t \uparrow \implies I_t \downarrow$
- * Therefore, the investment demand function is:

$$I_t = I_t^d(r_t^-, A_{t+1}^+, f_t^-, K_t^-)$$



- Household:

- In addition to consumption, household also enjoy leisure, which is simply defined by:

$$L_t = 1 - N_t$$

- Thus, the total utility is:

$$U = u(C_t, 1 - N_t) + \beta u(C_{t+1}, 1 - N_{t+1})$$

- Period t : Household earns labor income and dividends:

$$C_t + S_t = w_t N_t + D_t$$

- Period $t + 1$: Household also gains interest from savings and lending.

$$C_{t+1} = w_{t+1} N_{t+1} + D_{t+1} + D'_{t+1} + (1 + r_t) S_t$$

- Combining, we get the intertemporal budget constraint:

$$C_t + \frac{C_{t+1}}{1 + r_t} = w_t N_t + D_t + \frac{w_{t+1} N_{t+1} + D_{t+1} + D'_{t+1}}{1 + r_t}$$

- We can directly put the expression of C_{t+1} into the utility function to get an unconstrained maximization problem:

$$\max_{C_t, N_t, N_{t+1}} u(C_t, N_t) + \beta u((1 + r_t)(w_t N_t + D_t) + w_{t+1} N_{t+1} + D_{t+1} + D'_{t+1} - (1 + r_t)C_t, 1 - N_{t+1})$$

- Taking FOCs:

$$\begin{cases} \frac{\partial U}{\partial C_t} = u_C(C_t, 1 - N_t) - (1 + r_t)\beta u_C(C_{t+1}, 1 - N_{t+1}) = 0 \\ \frac{\partial U}{\partial N_t} = -(C_t, 1 - N_t) + (1 + r_t)\beta w_t u_C(C_{t+1}, 1 - N_{t+1}) = 0 \\ \frac{\partial U}{\partial N_{t+1}} = w_t \beta u_C(C_{t+1}, 1 - N_{t+1}) - \beta u_L(C_{t+1}, 1 - N_{t+1}) = 0 \end{cases}$$

Rearranging, we have:

$$\begin{cases} u_C(C_t, 1 - N_t) = (1 + r_t)\beta u_C(C_{t+1}, 1 - N_{t+1}) \\ u_L(C_t, 1 - N_t) = (1 + r_t)\beta w_t u_C(C_{t+1}, 1 - N_{t+1}) = w_t u_C(C_t, 1 - N_t) \\ u_L(C_{t+1}, 1 - N_{t+1}) = w_t u_C(C_{t+1}, 1 - N_{t+1}) \end{cases}$$

- The first equation is the Euler equation, which we have used to simplify the second one.
- Intuitions: MB = MC of consumption / leisure.
- Consumption demand is the same as before:

$$C_t = C^d(Y_t^+, Y_{t+1}^+, r_t^-)$$

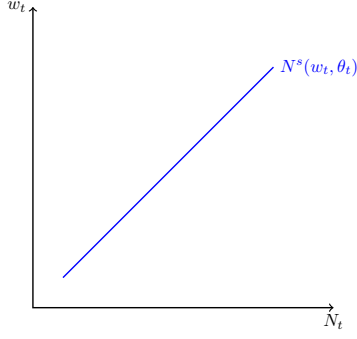
or if we have a government, by Ricardian equivalence:

$$C_t = C^d((Y_t - G_t)^+, (Y_{t+1} - G_{t+1})^+, r_t^-)$$

- The labor supply is more complicated to derive, so we use some assumptions.
- Specifically, we assume there is no income effect (wages increase, consume more, work less), but only substitution effect (work more because wages increase).
- Then, the labor supply function is:

$$N_t = N^s(w_t^+, \theta_t^-)$$

where θ_t represents how much one enjoys leisure.



– Example - GHH preferences: $u(C, 1 - N) = \ln(C + \theta \ln(1 - N))$

- Market-clearing conditions: Investment = Savings.

$$I_t = S_t$$

Then,

$$\begin{aligned} D'_{t+1} &= (r_t + f_t)I_t - r_t S_t \\ &= f_t I_t \end{aligned}$$

Also,

$$Y_t = w_t N_t + D_t = C_t + S_t = C_t + I_t$$

or adding government spending:

$$Y_t = C_t + I_t + G_t$$

6 Neoclassical RBC Model

- To sum up everything we have:

– Household:

$$\begin{aligned} C_t &= C^d(Y_t - G_t, Y_{t+1} - G_{t+1}, r_t) \\ N^t &= N^s(w_t, \theta_t) \end{aligned}$$

– Firm:

$$\begin{aligned} N^t &= N^d(w_t, A_t, K_t) \\ I_t &= I^d(r_t, A_{t+1}, f_t, K_t) \end{aligned}$$

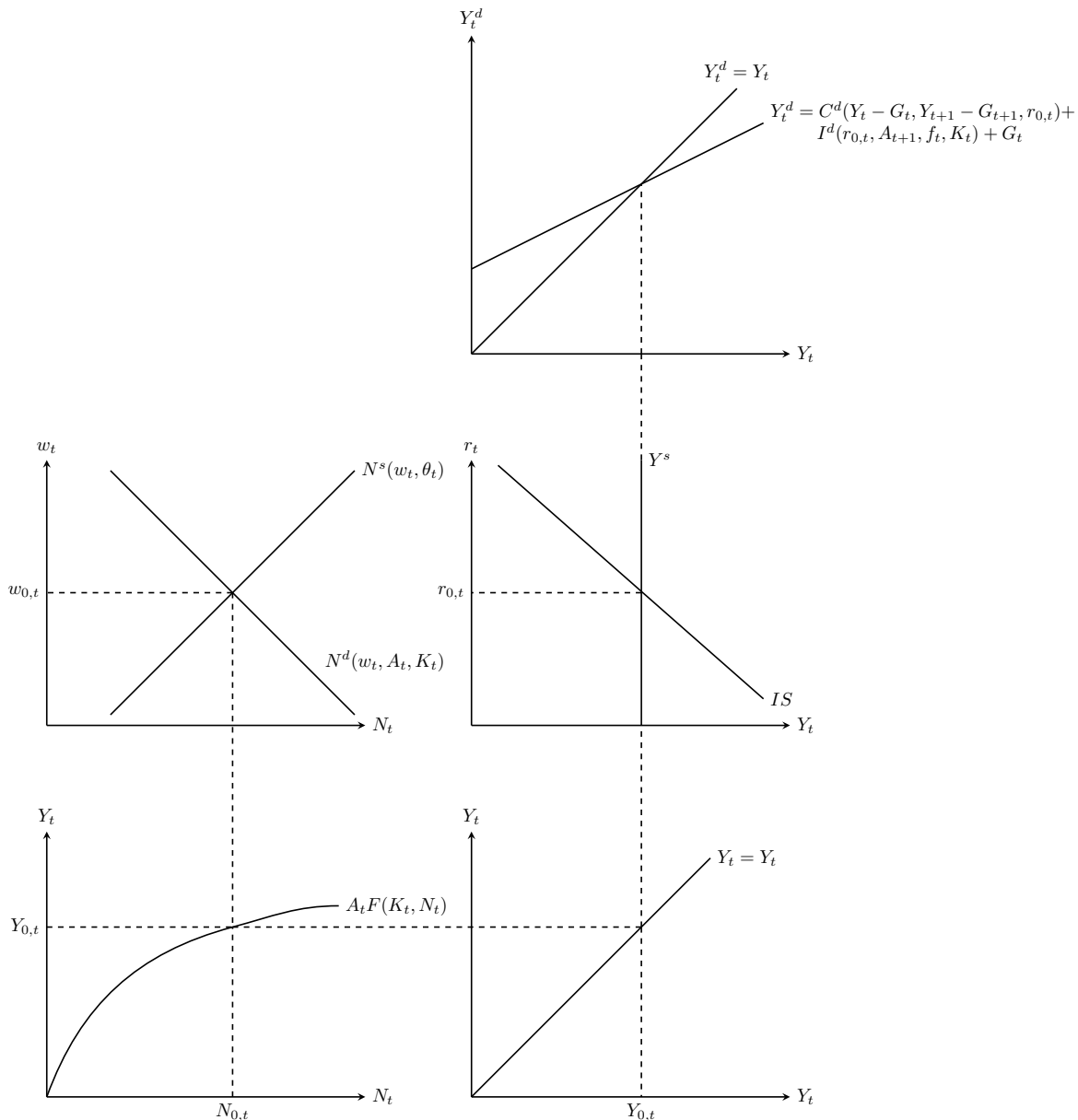
– Technological constraint:

$$Y_t = A_t F(K_t, N_t)$$

– Aggregate resources constraint:

$$Y_t = C_t + I_t + G_t$$

- Putting everything together, we can easily analyze the equilibrium graphically.



- We can graphically show the effect to the endogenous variables (w_t, N_t, Y_t, r_t) when any of the exogenous variable ($A_t, A_{t+1}, K_t, f_t, \theta_t, G_t, G_{t+1}$) changes.
- Most of the effects align with empirical observations.
 - More examples in the lecture notes.

7 Bonds and the Yield Curve

- We have been looking at the real economy with no money.
- Before moving on to introduce money, we first study the **nominal interest rate** dynamics, specifically the **yield curve**.
- For simplicity, we consider a two-period setting with a 1-year bond and a 2-year bond only.

- To invest \$1 for 2 years, an investor has two options:
 - Invest in the 1-year bond today and reinvest 1 year later.

$$\text{Expected return} = (1 + i_{1t})(1 + i_{1,t+1}^e)$$

- Invest in the 2-year bond directly.

$$\text{Expected return} = (1 + i_{2t})^2$$

- By the **expectation hypothesis**, these two should be exactly the same.

$$(1 + i_{2t})^2 = (1 + i_{1t})(1 + i_{1,t+1}^e)$$

Taking log approximation,

$$i_{2t} \approx \frac{i_{1t} + i_{1,t+1}^e}{2}$$

- If we include a risk premium x , the formula is simply:

$$i_{2t} \approx \frac{i_{1t} + i_{1,t+1}^e + x}{2}$$

- In this case, if $i_{1t} = i_{1,t+1}^e$, we still have $i_{2t} > i_{1t}$.
- This is why we often refer a upward-sloping yield curve as a "normal yield curve".
- If the yield curve is inverted:

$$i_{2t} < i_{1t} \implies i_{t+1}^e < i_t - x$$

meaning that people expect the interest rate to be much lower in the future, which often implies a negative economic outlook.

8 Money, Nominal Rigidities, AD-AS

- Money: medium of exchange, store of value, unit of account
- To introduce money, we study a money-in-utility model, where households are satisfied by holding money (for convenience in transactions).

$$u\left(C, \frac{M}{P}, 1 - N\right) = U(C) + \Gamma\left(\frac{M}{P}\right) + \theta V(1 - N)$$

- Example - Power utility: $U(C) = \frac{C^{1-\sigma}}{1-\sigma}$, $\Gamma\left(\frac{M}{P}\right) = \frac{\left(\frac{M}{P}\right)^{1-\gamma}}{1-\gamma}$.

- With money, we need to express the budget constraints in nominal terms.

- Period t :

$$P_t C_t + M_t + P_t S_t = P_t w_t N_t + P_t D_t$$

- Period $t + 1$: (We omit D'_{t+1} for simplicity)

$$P_{t+1} C_{t+1} + M_{t+1} + P_{t+1} S_{t+1} = P_{t+1} w_{t+1} N_{t+1} + P_{t+1} D_{t+1} + M_t + P_t(1 + i_t)S_t$$

- Terminal condition: $M_{t+1} = S_{t+1} = 0$.
 - Denote $\frac{M_t}{P_t}$ by m_t , we can express everything in real terms by dividing both sides by the price level:

- Period t :

$$C_t + m_t + S_t = w_t N_t + D_t$$

- Period $t + 1$:

$$C_{t+1} = w_{t+1} N_{t+1} + D_{t+1} + \frac{P_t}{P_{t+1}} m_t + \frac{P_t}{P_{t+1}} (1 + i_t) S_t$$

Using the fact that $\frac{P_{t+1}}{P_t} = 1 + \pi_{t+1}^e$ and the Fisher equation $1 + r_t = \frac{1+i_t}{1+\pi_{t+1}^e}$:

$$C_{t+1} = w_{t+1} N_{t+1} + D_{t+1} + \frac{m_t}{1 + \pi_{t+1}^e} + (1 + r_t) S_t$$

- Putting them together, we have the intertemporal budget constraint:

$$C_t + \frac{C_{t+1}}{1 + r_t} = w_t N_t + D_t + \frac{w_{t+1} N_{t+1} + D_{t+1}}{1 + r_t} - \left(\frac{i_t}{1 + i_t} \right) m_t$$

- Setting up the Lagrangian:

$$\begin{aligned} \mathcal{L} = & U(C_t) + \Gamma(m_t) + \theta V(1 - N_t) + \beta(U(C_{t+1}) + \theta V(1 - N_{t+1})) \\ & - \lambda \left(C_t + \frac{C_{t+1}}{1 + r_t} - w_t N_t - D_t - \frac{w_{t+1} N_{t+1} + D_{t+1}}{1 + r_t} + \left(\frac{i_t}{1 + i_t} \right) m_t \right) \end{aligned}$$

Taking FOCs:

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial C_t} = U'(C_t) - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial C_{t+1}} = \beta U'(C_{t+1}) - \frac{\lambda}{1 + r_t} = 0 \\ \frac{\partial \mathcal{L}}{\partial m_t} = \Gamma'(m_t) - \lambda \left(\frac{i_t}{1 + i_t} \right) = 0 \\ \frac{\partial \mathcal{L}}{\partial N_t} = -\theta_t V'(1 - N_t) + w_t \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial N_{t+1}} = -\beta \theta_t V'(1 - N_{t+1}) + \frac{w_t \lambda}{1 + r_t} = 0 \end{cases}$$

Rearranging,

$$\begin{cases} \lambda = U'(C_t) \end{cases} \quad (1)$$

$$\begin{cases} \lambda = \beta(1 + r_t) U'(C_{t+1}) \end{cases} \quad (2)$$

$$\begin{cases} \lambda = \left(\frac{1 + i_t}{i_t} \right) \Gamma'(m_t) \end{cases} \quad (3)$$

$$\begin{cases} \lambda = \left(\frac{\theta_t}{w_t} \right) V'(1 - N_t) \end{cases} \quad (4)$$

$$\begin{cases} \lambda = \beta(1 + r_t) \left(\frac{\theta_t}{w_t} \right) V'(1 - N_{t+1}) \end{cases} \quad (5)$$

- Now, we can look at different combinations of these equations:

- (1) and (2), (4) and (5) are the standard Euler equations.
- (1) and (4) gives us the labor supply:

$$\theta_t V'(1 - N_t) = w_t U'(C_t)$$

- We can derive the money demand from (1) and (3):

$$\frac{\Gamma'(m_t)}{U'(C_t)} = \frac{i_t}{1 + i_t}$$

Intuition: MRS between holding money and consumption = opportunity cost of holding money.

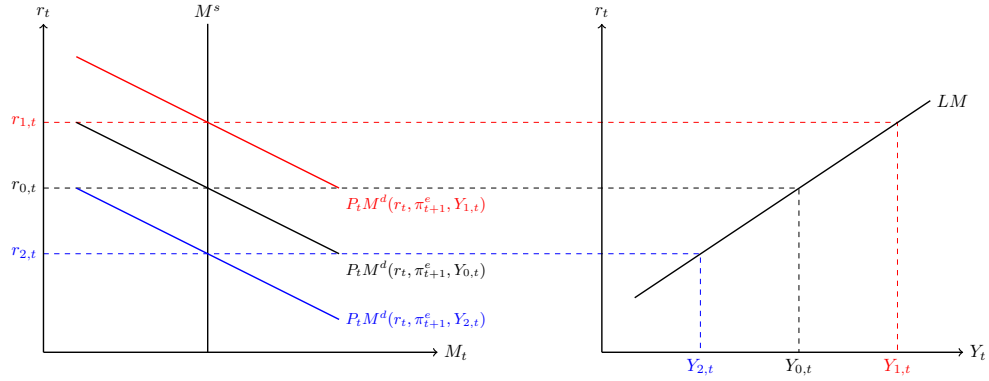
- In functional form (power utility):

$$\frac{M_t^d}{P_t} = \frac{M_t}{P_t} = C_t^{\frac{\sigma}{v}} \left(\frac{1 + i_t}{i_t} \right)^{\frac{1}{v}}$$

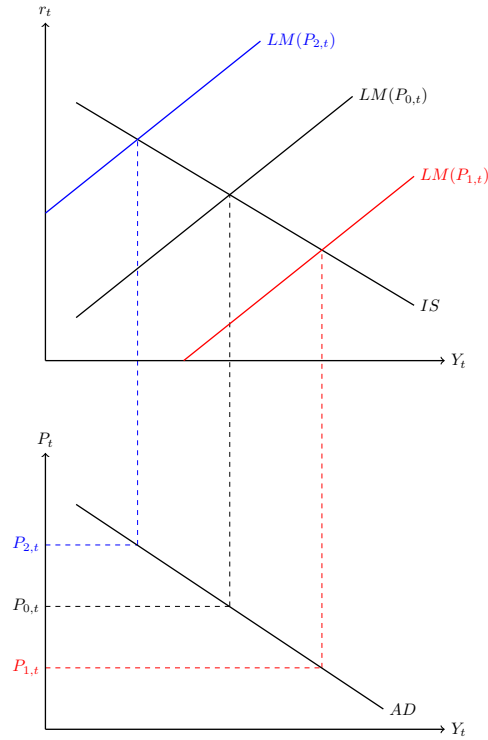
- Since C_t increases together with Y_t , we can write:

$$M_t = P_t M^d(i_t, Y_t) = P_t M^d((r_t + \pi_{t+1}^e)^-, Y_t^+)$$

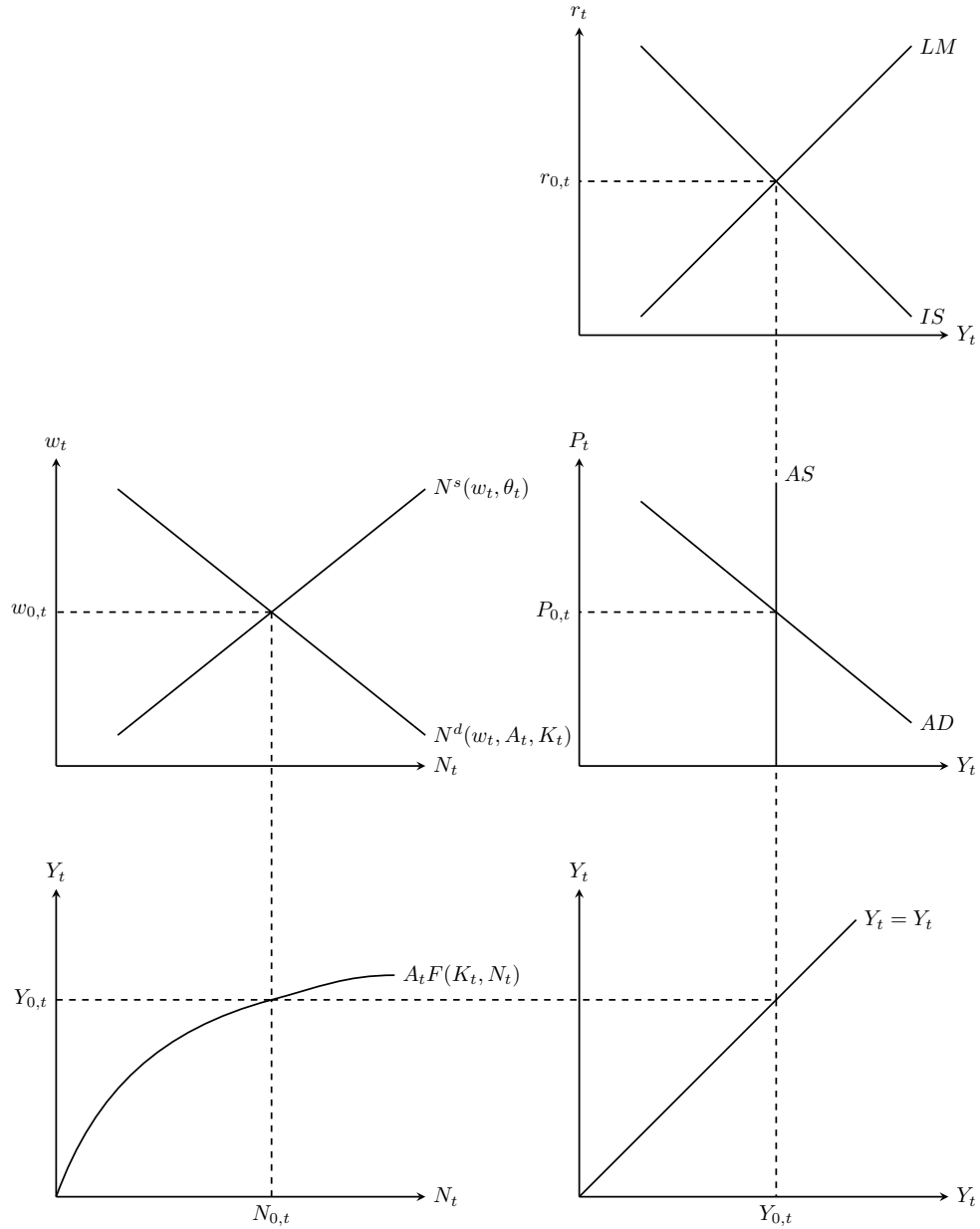
- Using these, we can derive the LM curve,



- and the AD curve.



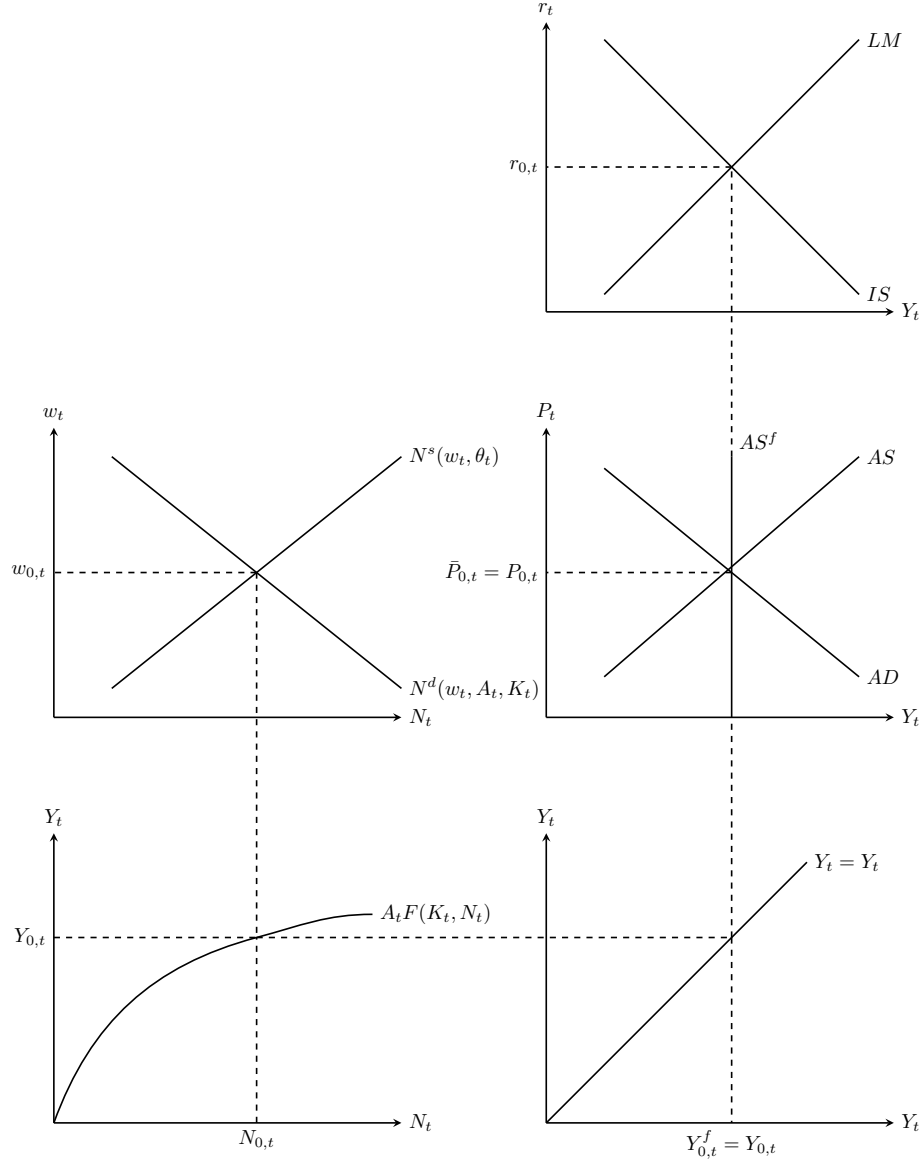
- The classical dichotomy: the real variables in the supply side (w_t, N_t , etc.) are unaffected by changes in the money supply or price levels.
- Putting everything together:



- Here, we can see that changes in Y_t must be driven by supply-side shocks, as Y_t is fixed by the vertical AS curve.
- A New Keynesian model makes use of a partial sticky price setting to the AS curve:

$$P_t = \bar{P}_t + \gamma(Y_t - Y_t^f)$$

- Therefore,



- Note that the full-employment AS curve, AS^f , is hypothetical here, and movements should occur along the upward-sloping AS curve.
- In this model, demand-side shocks usually cause inefficiency in the labor market.
- Again, examples of movements are omitted, but it is worth analyzing the effects of different shocks and fiscal / monetary policies.
- What if the central bank directly controls r_t ?

$$r_t = r((\ln Y_t - \ln Y_t^f)^+, P_t^+)$$

- Recall that:

$$\frac{M_t}{P_t} = C_t^{\frac{\sigma}{v}} \left(\frac{1 + r_t + \pi_{t+1}^e}{r_t + \pi_{t+1}^e} \right)^{\frac{1}{v}}$$

- Then the LM curve is still upward-sloping, but $M^s = \frac{M_t}{P_t}$ is now endogenous.